

Project Interim Report
On
Chat Application Simulator
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming with Java

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Abstract

- This project titled “**Chat Application Simulator**” is developed using Core Java and Object-Oriented Programming (OOP) concepts. The main aim of this project is to simulate a basic chat system where multiple users can communicate with each other in a structured manner.
- In today’s digital world, chat applications play a vital role in communication. This project is designed to understand how real-time messaging systems work internally. It allows users to join a chat room, send messages, and view the complete chat history in order. Each message contains details such as sender name, message content, and timestamp.
- The system uses classes such as User, Message, and ChatRoom to represent real-world entities. It demonstrates key OOP concepts like encapsulation, abstraction, and class relationships.
- This project helps in reducing manual communication complexity and provides a simple structured model of chat systems. It is useful for beginners to understand how messaging platforms function and how data is managed efficiently.

Introduction

- Chat applications have become an essential part of modern communication systems. Applications like WhatsApp and Messenger allow users to send and receive messages instantly. This project, “Chat Application Simulator,” is a simplified version of such systems designed using Java.
- The project focuses on simulating a chat environment where multiple users can interact within a chat room. It maintains message order and stores chat history, which can be displayed anytime.
- The system is built using Object-Oriented Programming concepts, making it modular and easy to understand. It consists of three main classes: User, Message, and ChatRoom.
- This project helps students understand how real-world communication systems are designed and implemented using programming principles.

Scope

This project is suitable for learning and small-scale chat system simulation. It can be extended in the future by adding:

- Database integration
- GUI interface
- Multiple chat rooms
- Real-time networking

System Requirements

Hardware:

- Computer/Laptop with minimum 4GB RAM

Software:

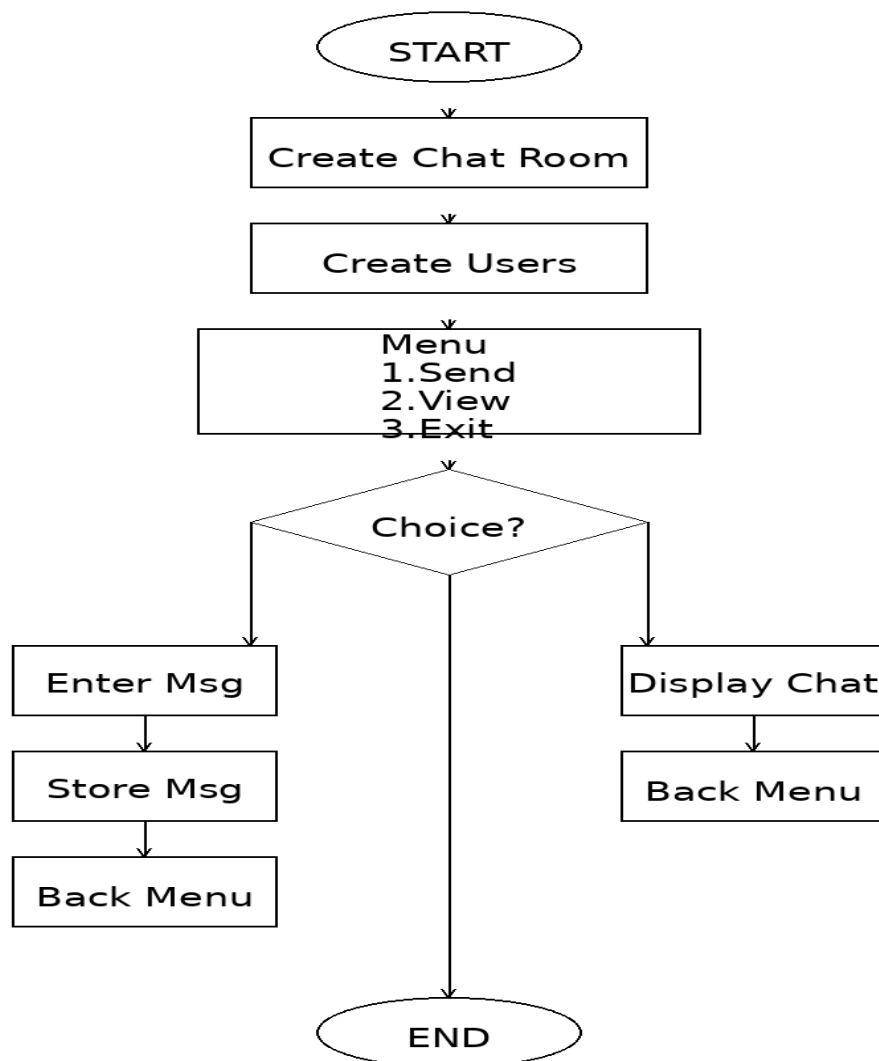
- Java JDK
- IDE (VS Code / Eclipse / IntelliJ)

Methodology

The working of the system follows these steps:

1. User enters the chat system
2. Users are created using the User class
3. Messages are created using the Message class
4. ChatRoom stores all messages in an array
5. When a user sends a message:
 - A Message object is created
 - It is stored in the array
6. The system maintains message order
7. Chat history is displayed by looping through messages

Flowchart



Implementation

The project is implemented using three main classes:

1. User Class

This class represents a user in the chat system. It stores the username and status of the user.

2. Message Class

This class represents a single message. It stores:

- Sender
- Message content
- Timestamp

3. ChatRoom Class

This is the main class that manages:

- Users
- Messages
- Chat history

It stores messages in an array and ensures they are displayed in correct order.

Gantt Chart

Planning & Structure

For the **Chat Application Simulator**, a Gantt chart helps divide the project into key tasks such as requirement analysis, UI design, backend development, message handling, and testing. Even if the application is rebuilt or modified, proper planning ensures smooth execution and better organization of tasks.

Time Management

The Gantt chart shows:

- When each phase of the chat application (design, coding, testing) begins
- When each phase is expected to be completed

This helps in tracking development progress, managing deadlines, and avoiding last-minute issues during implementation.

Conclusion

The **Chat Application Simulator** successfully demonstrates core Java Object-Oriented Programming (OOP) concepts such as encapsulation, inheritance, polymorphism, and abstraction. The system provides a functional simulation of real-time messaging, allowing users to send and receive messages efficiently. Overall, the project highlights both theoretical understanding and practical implementation of Java programming.

References

Book Format

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